Harjeet Singh Chahal

7325583249 | hc1242@scarletmail.rutgers.edu | linkedin.com/in/harjeet-singh-chahal | harjeet-chahal.github.io

Education

Rutgers University - New Brunswick, New Jersey, USA Aug 2025 - May 2027
M.S. in Computer Science **GPA: 4.00/4.00**

- Coursework: Machine Learning, Massive Data Mining, Interpretable & Explainable AI, Database Systems for Data Science, Data Structures & Algorithms, Artificial Intelligence, Visual Computing, Machine Vision.

Netaji Subhas University of Technology - New Delhi, India Nov 2020 - Jun 2024
B.Tech. in Computer Science & Engineering (Artificial Intelligence) **GPA: 3.74/4.00**

- Coursework: Deep Learning, Natural Language Processing, High Performance Computing, Distributed Computing, Operating Systems, Data Communication, Design and Analysis of Algorithms, Web Technology.

Technical Skills

Languages: Java, Python, C++, C, Go, SQL, JavaScript, TypeScript, HTML/CSS
Backend / Frontend: Node.js, FastAPI, Flask, Spring Boot, REST, gRPC, WebSockets, MCP, React, Tailwind CSS
Cloud & DevOps: AWS, GCP, Docker, Kubernetes, Terraform, Linux, Git, CI/CD, Github Actions, Prometheus, Grafana
Data Tools: PostgreSQL, pgvector, MySQL, MongoDB, Redis, ChromaDB, Kafka
AI / ML: PyTorch, Scikit-Learn, Hugging Face Transformers, NLTK, LangChain, OpenCV, NumPy, Pandas

Experience

Rutgers School of Health Professions Piscataway, NJ
Research Assistant Apr 2026 - Present

- Built and shipped a **Skin Health App** for iOS and Android using React Native, FastAPI, Docker and PostgreSQL.
- Deployed HIPAA aligned backend architecture on **Google Cloud** with server-side event logging for study analytics.
- Owned production infrastructure and release engineering, **delivering 15+ Apple TestFlight Builds for 100+ testers**.
- Integrated a context-aware Gemini chatbot, JWT auth, external APIs and CI/CD pipelines for automated deployment.

Technical Consulting and Research Inc. Weston, CT
Software Development Intern Jun 2026 - Aug 2026

- Orchestrated a **local-first cybersecurity compliance platform** (FastAPI, React, pgvector) running LLM inference and embeddings on-premise via Ollama, keeping client evidence off third-party APIs.
- Designed 8-criteria **LLM evaluation** rubric, architecting a RAG pipeline enforcing exact chunk-IDs to **stop hallucinations**.
- Developed an **MCP server** exposing 20 read-only tools for agentic retrieval, evidence lookup and control analysis.

Dhruv Research Gurugram, India
Associate Analyst Sep 2024 – May 2025

- Optimized Python and SQL pipelines to transform & aggregate 200k+ election survey responses across **15 state projects**.
- **Improved business-logic layers** for data-quality & validation workflows for demographic & geographic sampling in India.
- Collaborated with 8 member team through new quantitative metrics and reporting to **reduce manual work by 30%**.

Projects

FlexStore : AWS S3-Inspired Object Store | Go, gRPC, Docker | GitHub Sept 2025 – Dec 2025

- Built a distributed object store with stateless HTTP gateway & 3× replication across 5 nodes supporting chunked uploads.
- Engineered self-healing replication via **heartbeat failure detection** and replica failover, restoring durability in under 13s.
- Enforced SHA-256 integrity on each read/write & added **Prometheus metrics & Grafana dashboards** for observability.

SentryFlow : API Rate Limiting & Analytics | Python, Redis, Kafka | GitHub Jan 2026 – Mar 2026

- Architected an **API Gateway** in FastAPI sustaining 1500+ requests/s <100 ms latency feeding React analytics dashboard.
- Implemented **sliding-window & token-bucket algorithms** in Redis Lua, JWT auth & Kafka to ClickHouse ingestion.
- Containerized microservices with Docker & created Kubernetes ready system in Helm Chart maintaining 90% test coverage.

ScaleTrain : HPC Training Platform | PyTorch, DDP, NVIDIA A100 | GitHub May 2026 – June 2026

- Scaled a 30M-parameter GPT-transformer on Rutgers Amarel, reaching **500K tokens/s (2.66× speedup)** over baseline.
- Created SLURM fault-tolerant training with **signal-driven, atomic checkpoint/restart** and deterministic sharding, resuming from the exact interrupted step with zero lost work.

Achievements

Programming: Solved 1600+ Leetcode problems over 4 years consistently. | Leetcode
Academic Distinction: Ranked top 0.13% in India among ∼1.2M candidates in IIT JEE Main 2020; GRE Quant: 166/170.
Hackathon: Runner-Up, Rutgers Health Hack 2026 (Sevaro Track); Built a Tele-neurology operations platform. | Demo